GCE A LEVEL MARKING SCHEME

PREDICTED PAPER – ADVANCED

A LEVEL
ECONOMICS – COMPONENT 2
A520U20-1

About this marking scheme

The purpose of this marking scheme is to provide teachers, learners and other interested parties with an understanding of the assessment criteria used to assess this specific assessment.

This predicted (Advanced) marking scheme mirrors the structure, mark allocations and assessment objective weightings of the 2024 WJEC/Eduqas A Level Economics Component 2 paper. It is pitched at a demanding synoptic level, using the UK net-zero transition and the economics of digital platforms as its vehicles. It is not an official WJEC document.

GENERAL MARKING GUIDANCE

Positive Marking

It should be remembered that learners are writing under examination conditions and credit should be given for what the learner writes. It should be possible for a very good response to achieve full marks and a very poor one to achieve zero marks.

For each question there is a list of indicative content which suggests the range of economic concepts, theory, issues and arguments which might be included in learners' answers. This is not intended to be exhaustive and learners do not have to include all the indicative content to reach the highest level of the mark scheme.

The level-based mark schemes sub-divide the total mark to allocate to individual assessment objectives. Learners' responses to questions are assessed against the relevant individual assessment objectives and they may achieve different bands within a single question.

GCE A LEVEL ECONOMICS – COMPONENT 2

PREDICTED (ADVANCED) MARK SCHEME

Question 1

1	**1**	**Using the data, explain why UK greenhouse-gas emissions have fallen since 1990 (Figure 1).**	**[5]**

Band	AO1	AO2	AO3
2	**1 mark** **Knowledge** Clear knowledge of the drivers of emissions reduction.	**2 marks** **Good application** Strong reference to Figure 1 (-52% vs 1990) and Figure 2 (sectoral composition).	**2 marks** **Good analysis** Developed chain of reasoning explaining the fall.
1	**1 mark** **Limited** Some knowledge of emissions drivers.	**1 mark** **Limited** Limited references to the data.	**1 mark** **Limited** Chain of reasoning unconvincing or lacking in clarity.
0	**0 mark** No knowledge shown.	**0 mark** No valid application.	**0 mark** No valid analysis.

Indicative content

AO1

Emissions fall when the composition of energy supply shifts towards lower-carbon sources, when energy is used more efficiently, or when carbon-intensive activity is displaced abroad. Negative externality framework: carbon pricing and regulation shift the MPC curve closer to MSC.

AO2

Figure 1 shows UK territorial emissions fell from around 810 $MtCO_2e$ in 1990 to around 384 $MtCO_2e$ in 2023 – a reduction of about 52%. Figure 2 shows the sectoral distribution of remaining emissions: transport (28%), energy supply (20%), buildings (18%) and industry (14%). The biggest falls have been in the energy-supply sector, reflecting the phase-out of coal.

AO3

Decarbonisation of electricity generation: the "Dash for Gas" of the 1990s replaced coal with lower-carbon CCGTs; since the 2010s offshore wind has displaced gas. Coal's share of electricity has fallen from 70% in 1990 to under 2% in 2023.

The Climate Change Act 2008 set legally-binding carbon budgets, providing a policy framework that drove sustained investment. Carbon pricing (UK ETS, Carbon Price Floor) raised the marginal private cost of emitting, internalising the externality.

Energy-efficiency regulations (building standards, appliance standards, ESOS) reduced energy intensity per £ of GDP.

Deindustrialisation: a large part of the fall is attributable to the decline of heavy industry (steel, chemicals, cement) rather than genuine efficiency gains – some of the reduction is "offshored" emissions now produced overseas for UK consumption.

Renewables cost reductions: LCOE of offshore wind has fallen over 70% since 2010, making low-carbon alternatives economically attractive.

Transport changes have been slower (still 28% of emissions) – the vehicle fleet is cleaner but growth in traffic has offset some gains.

Allow any other valid points.

1	2	**Using the data, explain why the UK Government might want to accelerate investment in low-carbon technologies (Figure 3).**	**[5]**

Band	AO2	AO3
		3 marks **Excellent analysis** Well-developed reasoning covering multiple rationales for acceleration.
2	**2 marks** **Good application** Strong use of Figures 3 and 5.	**2 marks** **Good analysis** Well-developed chains of reasoning.
1	**1 mark** **Limited** Use of data limited.	**1 mark** **Limited** Points under-developed.
0	**0 mark** None.	**0 mark** None.

Indicative content

AO2

Figure 3 shows UK low-carbon investment of around £26 billion in 2023, below the £50-70 billion per year required by the Climate Change Committee. Figure 5 shows LCREE employment reaching 432,000 in 2023, up from around 250,000 in 2015. Figure 4 shows the UK's emissions-per-capita is already below the US and Germany. The US Inflation Reduction Act offers around $370 billion in subsidies for clean technology.

AO3

Meeting legally-binding targets: the Climate Change Act 2008 commits the UK to a 68% reduction by 2030 and Net Zero by 2050. Failure to invest at a faster rate makes those targets unreachable.

First-mover advantage: the UK offshore-wind sector leveraged early investment to become a world leader. Extending this to heat pumps, CCUS and green hydrogen could build export industries and long-term comparative advantage.

Negative externality: emissions impose significant external costs (climate damage, air-pollution health costs); accelerating low-carbon investment internalises the externality and moves towards MSC = MSB.

Green jobs: LCREE employment is growing faster than the economy overall – investment generates high-skilled, often regional employment, supporting "levelling up".

Energy security: domestic renewables reduce exposure to volatile global gas markets (as the 2022 price shock demonstrated).

International competition: the US IRA and EU Green Deal risk diverting investment away from the UK unless the UK matches the policy ambition.

Public finance: Net Zero investment is front-loaded but delivers long-term returns (energy security, avoided damage, productivity). Delay raises the total cost of the transition.

Allow any other valid points.

1	3	How effective is the UK's net-zero policy framework (Figure 8) likely to be in delivering the transition to a low-carbon economy?	[10]

Band	AO2 (3 marks)	AO3 (3 marks)	AO4 (4 marks)
3	**3 marks** **Excellent application** Data used on both sides, linking Figure 8 to Figures 1, 3 and 6.	**3 marks** **Excellent analysis** Well-developed chains explaining how each policy element drives decarbonisation.	**3-4 marks** **Excellent evaluation** Well-developed counter-arguments on weaknesses of the framework. Top-band answers give a well-reasoned overall judgement.
2	**2 marks** **Good** Data well used on one side of the argument.	**2 marks** **Good** Developed chains showing how the framework works.	**2 marks** **Good** Developed counter-arguments on weaknesses.
1	**1 mark** **Limited** Use of data less effective.	**1 mark** **Limited** Chains of reasoning under-developed.	**1 mark** **Limited** Counter-arguments present but under-developed.
0	**0 marks** None.	**0 marks** None.	**0 marks** None.

Indicative content

Effectiveness (AO3)

Carbon pricing (UK ETS, Carbon Price Floor) internalises the externality of emissions. By raising MPC closer to MSC, polluters are incentivised to invest in abatement where it is cheapest – allocative-efficiency gain. The falling cap on permits guarantees emissions fall over time.

CCUS and hydrogen investment (£20bn cluster programme) addresses the market failure of under-investment in new technology – private firms cannot capture the full social benefit of R&D, so state co-funding corrects this.

Demand-side subsidies (Boiler Upgrade Scheme £7,500, EV grants) reduce the upfront cost barrier for households (Figure 6).

Sector-specific regulations (phase-out of new ICE cars by 2035, boiler standards) provide certainty to manufacturers and accelerate adoption through mandates.

CBAM (EU and prospective UK) addresses carbon leakage – the risk that UK emissions reductions are offset by imports from unregulated jurisdictions.

Institutional framework (CCC, statutory budgets) provides credibility and prevents policy reversal – a powerful anchor for long-term private investment.

Weaknesses (AO4)

The 2023 rollbacks (five-year delay to ICE ban, weakening of boiler phase-out) have undermined policy credibility. The CCC warned in 2024 that current policies are insufficient to meet the 2030 target.

Investment (Figure 3) of £26bn is well below the £50-70bn the CCC estimates is required, implying the framework is inadequately scaled.

Regressive distributional impacts: carbon prices raise energy and goods prices, hitting low-income households hardest. Figure 6 shows low-carbon technologies are 2-4 times more expensive upfront.

International competition: the US IRA (worth $370bn) is diverting green investment to the US. Without similar scale, the UK framework may fail to attract firms.

Technology bets: CCUS remains unproven at scale; over-reliance on this technology may delay more certain action.

Grid constraints: even if investment accelerates, grid bottlenecks limit the speed of connection for new renewables – a binding constraint the framework doesn't fully address.

Free-rider problem: UK emissions are only around 1% of global total. Unilateral action cannot stabilise the climate.

Overall: the framework combines the right conceptual elements (price signal + targeted support + regulation + institutions) but is seriously under-scaled relative to the target, with patchy political commitment.

Allow any other valid points.

1	4	**Using the data, discuss the extent to which the distribution of green jobs across UK regions (Figure 7) is likely to reduce UK regional inequalities.**	**[10]**

Band	AO2 (3 marks)	AO3 (3 marks)	AO4 (4 marks)
3	**3 marks** **Excellent** Figure 7 used in detail on both sides.	**3 marks** **Excellent** Well-developed chains on how green-jobs distribution could reduce inequality.	**3-4 marks** **Excellent** Well-developed counter-arguments and a well-reasoned overall judgement.
2	**2 marks** **Good** Figure 7 used on one side of the argument.	**2 marks** **Good** Developed chains of reasoning.	**2 marks** **Good** Developed counter-arguments.
1	**1 mark** **Limited** Use of Figure 7 superficial.	**1 mark** **Limited** Chains under-developed.	**1 mark** **Limited** Counter-arguments under-developed.
0	**0 marks** None.	**0 marks** None.	**0 marks** None.

Indicative content

Reduces inequality (AO3)

Figure 7 shows Scotland projected to gain +32,000 jobs and the North East +18,000 – both regions with below-average per-capita income. Large-scale offshore wind (Dogger Bank, Hornsea) and hydrogen clusters are concentrated in former industrial areas.

Yorkshire & Humber benefits from the Humber CCUS cluster and retrofitted industrial capacity, replacing jobs lost in coal and heavy industry.

Green jobs offer higher wages than average regional employment (LCREE average wage ~£38k vs regional averages often £28-30k), so per-worker earnings rise faster in these regions.

Infrastructure investment (grid, ports, hydrogen pipelines) generates local multipliers – construction jobs, supply-chain firms, local services – amplifying the income impact.

Skills matter: many green jobs leverage traditional engineering skills (offshore oil & gas workers transitioning to offshore wind), mitigating the transition cost for declining sectors.

Regional industrial strategies (Freeports, Investment Zones) layered on top of green investment mean lagging regions capture more of the new economic activity.

Limits to impact (AO4)

Figure 7 also shows the South East (+20,000) and London (+14,000) capturing significant green jobs (R&D, finance, engineering headquarters). The absolute number may not close the regional gap if the richer South captures the high-value-added functions.

Per capita matters: Scotland's population (5.5m) and the North East's (2.6m) differ significantly. +32,000 jobs in a small region is more transformational than +45,000 in a larger one.

Job quality varies: some offshore-wind manufacturing jobs have been lost overseas. If the UK captures only installation and maintenance, value-added stays lower.

Geographic concentration risk: offshore wind jobs are coastal; inland towns (Stoke-on-Trent, Burnley) may miss out entirely – regional inequality could simply shift rather than narrow.

Skills gap: specialist green jobs require new training; existing workers in declining sectors may not access them without significant retraining support.

Displacement: some "new" green jobs displace fossil-fuel jobs in the same regions. If existing fossil workers cannot transition, regional employment impact is muted.

Other levers matter more: regional inequality is driven by productivity, skills, connectivity and public investment. Green jobs are a useful contribution but not a complete solution.

Overall judgement: the distribution of green jobs is mildly progressive in regional terms (bigger absolute gains in the North East, Scotland, Yorkshire), but it is not a silver bullet. Success depends on capturing high-value activities, training local workers, and complementary regional investment.

Allow any other valid points.

1	5	Using the data, discuss whether the higher upfront cost of low-carbon technologies (Figure 6) is likely to make the net-zero transition regressive.	[10]

Band	AO2 (3 marks)	AO3 (3 marks)	AO4 (4 marks)
3	**3 marks** **Excellent** Figure 6 used in detail, with reference to at least two technologies.	**3 marks** **Excellent** Well-developed chains on the regressive impact.	**3-4 marks** **Excellent** Well-developed counter-arguments and overall judgement.
2	**2 marks** **Good** Figure 6 used on one side.	**2 marks** **Good** Developed chains.	**2 marks** **Good** Developed counter-arguments.
1	**1 mark** **Limited** Superficial use of Figure 6.	**1 mark** **Limited** Chains under-developed.	**1 mark** **Limited** Counter-arguments under-developed.
0	**0 marks** None.	**0 marks** None.	**0 marks** None.

Indicative content

Regressive impact (AO3)

Figure 6 shows a heat pump costs £8,000-13,000 versus £2,000-3,500 for a gas boiler – an upfront gap of £5,000-10,000 that is unaffordable for the 30% of UK households with less than £1,000 in savings.

EVs typically cost £30,000-40,000+ versus £20,000-25,000 for an equivalent ICE car; low-income households purchase second-hand only, where the EV market is smaller and more expensive.

Solar PV (£6,000-10,000) is only practical for owner-occupiers – ruling out 36% of UK households who rent and who cannot recoup the investment before moving.

Insulation measures have significant upfront costs but benefit mainly lower-income households living in poorly-insulated homes; without subsidy they are priced out.

Carbon pricing passed through to bills is regressive: low-income households spend a higher share of income on energy, fuel and food.

Residualisation: if only wealthy households switch to heat pumps, the remaining gas network has fewer users sharing its fixed costs, raising bills for those left behind.

Mitigation / not regressive (AO4)

Running costs favour low-carbon technologies: EVs cost around 3p/mile versus 12p/mile for petrol; heat pumps deliver 300% efficiency versus 90% for gas boilers. Over the lifetime, total cost of ownership can be lower.

Subsidies: the Boiler Upgrade Scheme offers £7,500 off a heat pump, significantly closing the gap. The Warm Homes Plan funds insulation for low-income households.

Social tariffs: targeted energy support has been used during price shocks and could be extended as a structural policy tool.

Second-hand EV market is growing rapidly: as the new market matures, prices will fall and access improves.

Offshore-wind strike prices as low as £44/MWh are already below gas-CCGT prices – as the generation mix decarbonises, wholesale electricity prices may fall, with the biggest relative benefits going to those who spend most on energy.

Carbon dividends: if carbon-pricing revenues are recycled as per-capita dividends (as in Canada), the scheme can be strongly progressive.

Mandates and regulation can shift costs to manufacturers and developers (e.g. new-build standards), protecting existing owners from compulsory retrofit costs.

Public infrastructure (district heating, public EV charging) can deliver low-carbon services without household-level capex.

Overall: the upfront cost profile is undeniably regressive in the short run without policy correction. But subsidy design, running-cost advantages over time, and revenue-recycling mean the transition can be made progressive on balance. Whether it is depends on political choices about redistribution.

Allow any other valid points.

Question 2

2	1	**Using the information on digital platforms, outline the characteristics of a market with substantial network effects and high fixed costs.**	**[5]**

Band	AO1	AO2
3		**3 marks** **Excellent application** Strong use of the case to illustrate network effects and high-fixed-cost features.
2	**2 marks** **Good knowledge** Comprehensive understanding of network-effect markets.	**2 marks** **Good application** Good use of the case to illustrate one concept well.
1	**1 mark** **Limited** Understanding superficial or incomplete.	**1 mark** **Limited** References to the market are superficial.
0	**0 marks** No knowledge shown.	**0 marks** No valid application.

Indicative content

AO1/AO2

Direct network effects: the value of the service to each user rises as more users join (e.g. WhatsApp, Facebook). A user joining Meta's platforms adds value for every existing user.

Indirect (two-sided) network effects: platforms connect distinct user groups (e.g. buyers and sellers, developers and app-users). Apple's App Store and Google Play are two-sided: more users attract more developers, which attracts more users.

Winner-takes-most dynamics: once a platform reaches a critical mass it becomes self-reinforcing and dominance tends to persist. Figure 1 shows Google with ~92% of search, iOS + Android with ~99% of mobile operating systems.

High fixed costs, very low marginal costs: platforms require massive investment in data centres, engineering and R&D ($225bn in 2022 for the top five) but the marginal cost of serving an additional user is close to zero – similar to natural-monopoly cost structures.

Data moats: accumulated user data feeds into superior recommendations, search results and advertising, which attracts more users and more data – a feedback loop that raises entry barriers.

Switching costs and lock-in: users' data, contacts, photos, purchase histories are held in proprietary formats; leaving a platform imposes significant cost.

Zero-price business models: consumers "pay" with attention and data; revenue is extracted from advertisers or from one side of a two-sided market (e.g. 30% app-store commission on developers).

Platform envelopment: a platform dominant in one market can leverage it to enter adjacent markets (e.g. Apple extending from devices to services).

2	2	**Using Figure 2, calculate the marginal revenue (MR) and marginal cost (MC) for each of the subscriber levels (0 to 800,000). Record the answers in your pink answer booklet. Using your answers explain at what level of output PulseApp Ltd will, in theory, maximise profits.**	**[7]**

Band	AO1	AO2	AO3
3		**3 marks** **Excellent** Comprehensive display of all MC and MR data.	**3 marks** **Excellent** Well-developed chain of reasoning explaining the profit-maximising output.
2		**2 marks** **Good** Strong use of data with either MC or MR correctly calculated.	**2 marks** **Good** Developed chain showing how profit maximisation is determined.
1	**1 mark** **Knowledge** Knowledge of the profit-maximisation condition.	**1 mark** **Limited** Use of data limited with significant inaccuracies.	**1 mark** **Limited** Chain of reasoning unconvincing.
0	**0 mark** No knowledge.	**0 mark** No valid application.	**0 mark** No valid analysis.

Indicative content

AO1

Profit maximisation occurs where MC = MR (or the highest output at which MR ≥ MC).

AO2

Price (£/month)	Subs (000s)	TR (£000s)	MR (£000s per 000 subs)	TC (£000s)	MC (£000s per 000 subs)
£18	0	0	–	500	–
£16	100	1,600	16	520	0.2
£14	200	2,800	12	560	0.4
£12	300	3,600	8	620	0.6
£10	400	4,000	4	720	1.0
£8	500	4,000	0	860	1.4
£6	600	3,600	−4	1,060	2.0
£4	700	2,800	−8	1,340	2.8
£2	800	1,600	−12	1,720	3.8

For MR and MC both correctly calculated: AO2: 3. For MR and MC correctly calculated with minor errors: AO2: 2. For one of MR or MC correctly calculated: AO2: 1.

AO3

Profits are maximised at 400,000 subscribers where MR (4) comfortably exceeds MC (1.0). At 500,000 subscribers, MR (0) < MC (1.4), so that increment reduces profit. Total profit (TR − TC) at 400,000 = £4,000k − £720k = £3,280k (£3.28 million).

Candidates should identify that this market exhibits the classic digital-platform cost profile: very high fixed costs (£500k even at zero output) with very low marginal costs – which is why the profit-maximising price (£10) remains well above MC. Allow OFR.

2	**3**	**Using a costs and revenue diagram, evaluate the effects on a dominant digital platform's abnormal profits of the imposition of regulation requiring it to offer interoperability with rival platforms.**	**[10]**

Band	AO1 (3 marks)	AO2 (2 marks)	AO3 (2 marks)	AO4 (3 marks)
3	**3 marks** **Excellent** Accurate diagram showing initial equilibrium and shift following interoperability.			**3 marks** **Excellent** Impact on abnormal profits fully discussed with a judgement.
2	**2 marks** **Good** Diagram with some minor errors.	**2 marks** **Good** Case used well (DMA, CMA mobile ecosystems, EU precedent).	**2 marks** **Good** Chain explaining why abnormal profits fall.	**2 marks** **Good** Well-developed counter-arguments.
1	**1 mark** **Limited** Diagram has significant errors.	**1 mark** **Limited** Case used weakly.	**1 mark** **Limited** Chain less well-developed.	**1 mark** **Limited** Counter-arguments lack development.
0	**0 mark** None.	**0 mark** None.	**0 mark** None.	**0 mark** None.

Indicative content

AO1

Diagram shows the initial monopoly/oligopoly equilibrium with output where MC = MR, price set on AR above AC – the shaded rectangle between AR and AC at Q* is abnormal profit (1 mark).

Interoperability reduces user lock-in; AR and MR shift leftwards because existing users can now access rival services (lower market share at any price) (1 mark).

AC may shift up slightly to reflect compliance/engineering costs of implementing interoperability APIs; MC broadly unchanged. New equilibrium at lower Q* with smaller abnormal-profit rectangle (1 mark).

AO2

Apple/Google currently charge 30% commission (case) – interoperability would let developers route payments outside the app-store, directly reducing this revenue stream.

The EU's Digital Markets Act (DMA) already requires interoperability of messaging for designated gatekeepers – a direct case example.

The CMA's mobile-ecosystems market investigation (referenced) proposes similar remedies for the UK.

Google's $18bn default-search payment to Apple is vulnerable if interoperability reduces switching costs and regulators question default-search deals.

AO3

Network effects are the main competitive moat for platforms. Interoperability reduces the "stickiness" of the network: a user of platform A can now interact with users of platform B. The strategic complementarity that sustained dominance is weakened; AR shifts left.

Elasticity of demand rises (switching becomes easier) so the platform's mark-up over marginal cost falls. Both price and quantity may fall, reducing TR.

Abnormal profits shrink: in the extreme (perfect interoperability + no other barriers), competition is restored and profits fall towards the normal level.

Compliance costs (engineering APIs, security) raise AC.

AO4

Depends on how interoperability is designed: mandated message interoperability (WhatsApp ↔ Signal) may have only limited effect on revenue because advertising revenue is driven by attention, not lock-in per se.

Platforms may respond by differentiating the user experience, investing in quality, and retaining users through non-network features – innovation response can preserve profits.

Compliance costs may be significant: Apple's Vision Pro and iMessage engineering demonstrate how costly interoperability retrofits can be.

Reduced abnormal profits may reduce incentive to invest in platform R&D – the case notes $225bn of R&D by the top five firms.

Developer-side effects: if commission rates fall from 30% to say 15%, app-store revenue falls sharply – but new apps may enter, increasing platform usage, partially offsetting revenue loss.

Experience from EU DMA: preliminary evidence suggests designated gatekeepers have responded with substantial investment in compliance but continued strong revenue growth.

First-order effect: abnormal profits fall. Second-order effects (innovation, platform quality, consumer response) may moderate or amplify the result.

Overall: interoperability regulation is likely to reduce abnormal profits in the short run, but the magnitude depends on design detail, scope, and platform response.

Allow any other valid points.

2	4	**Evaluate the view that competition authorities should break up dominant digital platforms in order to improve market outcomes.**	**[7]**

Band	AO1	AO3	AO4
3			**3 marks** **Excellent** Strong evaluation judging whether breakup is the right remedy, with a clear overall position.
2	**2 marks** **Good** Clear understanding of breakup, monopoly and market outcomes.	**2 marks** **Good** Clear chain of reasoning on why breakup could improve welfare.	**2 marks** **Good** Developed counter-arguments on why breakup may be undesirable.
1	**1 mark** **Limited** Partial understanding shown.	**1 mark** **Limited** Chain lacks clarity or detail.	**1 mark** **Limited** Counter-arguments under-developed.
0	**0 marks** None.	**0 marks** None.	**0 marks** None.

Indicative content

AO1

Breakup (structural separation) involves dividing a dominant firm into independent entities, typically by business line (e.g. separating a platform's app-store from its operating system) or geographically. Classic precedents are the 1911 Standard Oil breakup and the 1982 AT&T breakup. Competition authorities (CMA, FTC, European Commission) have the power to order structural remedies where behavioural remedies are inadequate.

In favour of breakup (AO3)

Addresses monopoly pricing: Apple and Google's 30% app-store commission represents monopoly rent extraction from developers; separating the app store from the operating system would expose it to competition.

Restores competition in search: Google's $18bn payment to Apple entrenches dominance; a structural remedy could allow rivals genuine access.

Reduces data concentration: separating Instagram, WhatsApp and Facebook would limit Meta's ability to use cross-platform data for advertising.

Historical precedent: AT&T breakup led to a decade of rapid innovation in telecoms. Standard Oil breakup preserved a competitive oil-refining sector.

Simpler to enforce than behavioural remedies: once separated, ongoing compliance monitoring is lighter.

Re-opens markets to new entrants, addressing the winner-takes-most dynamic.

Against breakup (AO4)

Loss of network effects reduces user value: users chose Meta's platforms partly because their friends were there. A forced breakup may deliver a worse user experience.

Innovation impact: the case notes $225bn of R&D spend by the top five firms, largely funded by platform rents. Breakup risks slowing innovation.

"Zero-price" consumer benefits: consumers currently receive search, social, email at zero money cost. Breakup may force consumers to pay directly.

Conglomerate efficiencies: shared services (cloud, identity, payment systems, security) deliver economies of scope. Breakup may raise AC across the separated businesses.

Defining the relevant market is extremely difficult for multi-sided platforms.

Global competitiveness: Chinese platforms (Alibaba, ByteDance, Tencent) are already large; US/UK breakups risk weakening domestic champions without affecting global competition.

Less-drastic alternatives exist: the DMA and the CMA's conduct requirements for SMS firms can deliver many of the competition benefits without the structural costs. Interoperability, data-portability and non-discrimination rules are targeted and reversible.

Implementation risk: breakup is slow (Microsoft breakup was reversed on appeal; FTC's Meta case ongoing after four years).

Overall: breakup is a powerful tool but a blunt one. The DMA/CMA regulatory approach should be given time to work; breakup remains a long-stop option.

Allow any other valid points.

2	5	**With reference to the data, discuss the extent to which dominant digital platforms are harmful to economic welfare.**	**[11]**

Band	AO1 (2 marks)	AO2 (2 marks)	AO3 (3 marks)	AO4 (4 marks)
3			**3 marks** **Excellent** Well-developed explanation of how platforms harm welfare.	**3-4 marks** **Excellent** Very well-developed counter-arguments on consumer benefits and innovation, with a well-reasoned overall judgement.
2	**2 marks** **Good** Clear understanding of economic welfare.	**2 marks** **Good** Well-contextualised use of the case on both sides.	**2 marks** **Good** Developed explanation of welfare effects.	**2 marks** **Good** Developed counter-arguments.
1	**1 mark** **Limited** Some knowledge of welfare.	**1 mark** **Limited** Some references, not well developed.	**1 mark** **Limited** Partial analysis.	**1 mark** **Limited** Counter-arguments under-developed.
0	**0 marks** None.	**0 marks** None.	**0 marks** None.	**0 marks** None.

Indicative content

AO1

Economic welfare is the sum of consumer and producer surplus, adjusted for externalities. It can be assessed statically (allocative and productive efficiency) and dynamically (innovation, new products). Market failures arise from monopoly power, externalities and information asymmetry.

AO2

Apple and Google collect ~30% app-store commission (case) – a direct extraction from developer surplus.

Google pays Apple $18bn/year to be the default search engine (case) – a payment to suppress search competition.

Top five firms spent $225bn on R&D in 2022 – around 14% of revenue – producing substantial innovation (case).

Many platform services (search, social, email) are zero-price to users (case) – a large consumer-surplus benefit.

Figure 1 market shares: Google 92% of search, iOS + Android 99% of mobile OS – extreme concentration.

Harm to welfare (AO3)

Monopoly pricing in B2B markets: 30% app-store commission extracts developer surplus and, because many developers pass the cost on, extracts consumer surplus too – a classic deadweight loss from monopoly pricing.

Barriers to entry through default deals ($18bn Google-Apple): rivals with better products cannot get access to consumers because distribution is bought up. Allocative inefficiency.

Data exploitation and information asymmetry: users do not fully understand what data is collected or how it is used; platforms use the data to target advertising that may manipulate behaviour, exploiting bounded rationality.

Suppression of rivals through platform envelopment: Meta's integration of Instagram, WhatsApp and Messenger reduces choice; Amazon's Basics line competes with marketplace sellers using their own data.

Innovation chilling: smaller firms that threaten incumbents can be acquired at the "kill-zone" stage, reducing long-run innovation.

Negative externalities: social-media business models have been linked to mental-health harms in teenagers (evidence contested but growing), and to political-polarisation costs that are external to the platform's profit calculation.

Benefits / limits to harm (AO4)

Massive consumer surplus from zero-price services: the estimated value to US consumers of free search alone is in the hundreds of billions per year (Brynjolfsson); a huge welfare gain.

R&D and innovation: $225bn of R&D has delivered smartphones, cloud computing, voice assistants, mapping, translation, and is now delivering generative AI. Much of this would not exist without the platform business model.

Network effects genuinely raise user value: a social network with one user is worthless; the economic value comes from scale. Breaking up or shrinking platforms destroys value for users, not just shareholders.

Economies of scale: the very high fixed costs of global platforms can only be recovered by operating at massive scale.

Global competitiveness: US and UK/EU platforms provide strategic capabilities that would otherwise be supplied by Chinese or other foreign firms with their own governance concerns.

Regulation (DMA, CMA SMS regime) already addresses many of the harms – we should evaluate welfare under the regulated equilibrium.

Developer welfare: while commissions are high, many developers would not exist as businesses without the app-store reach the platforms provide.

Bidirectional network effects benefit consumers: more developers = more apps = better consumer experience.

Harm is concentrated in B2B (app developers, advertisers, sellers) rather than directly in consumer-facing services.

Overall judgement: the welfare impact is mixed. Consumer surplus gains from zero-price innovation are vast but imperfect (data costs, externalities). Harm is real in B2B markets and in indirect ways through innovation suppression and default deals. A targeted regulatory response is more likely to raise welfare than breakup; the balance of evidence suggests regulated dominance rather than structural remedies delivers the best welfare outcome – but this depends on effective enforcement.

Allow any other valid points.

Assessment Objectives Grid

	AO1	AO2	AO3	AO4	Total	No. parts
1a	1	2	2		5	2
1b		2	3		5	2
1c		3	3	4	10	2
1d		3	3	4	10	2
1e		3	3	4	10	2
2a	2	3			5	2
2b	1	3	3		7	2
2c	3	2	2	3	10	3
2d	2		2	3	7	2
2e	2	2	3	4	11	3
Total	**11**	**23**	**24**	**22**	**80**	

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